

Comparative Evaluation of Deep Audio Models for Detecting AI Generated Music

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<https://github.com/minahilali117/Synthetic-Music-Detector>

Abstract. Concerns about authenticity, provenance, and abuse in digital media ecosystems have grown as AI generated music has quickly improved in fidelity and diversity. As a result, identifying AI generated audio has become a crucial technical task with wide ranging effects for copyright, cultural integrity, and legal compliance. In this work, we present a comparative study of several deep learning-based classifiers intended to differentiate between audio written by humans and music produced by artificial intelligence. Four main model families are evaluated in this study: a hybrid model that combines latent features from the AE and transformer embeddings, a human-trained autoencoder (AE) representation model, a transformer-based audio representation model, and a convolutional neural network (CNN) that operates directly on mel-spectrograms.

This paper makes three contributions: (1) we design a unified preprocessing and representation pipeline for multi-source human and AI audio datasets; (2) we perform a thorough comparative evaluation using ROC-AUC, accuracy, F1-score, and qualitative embedding-space visualizations; and (3) we offer an analysis framework that addresses dataset bias, generalization, and the limitations of existing detection systems. Sections that need to be inserted with updated findings from the final Transformer model are clearly noted.

Keywords: AI generated music · audio classification · representation learning · transformer models · hybrid models

1 Introduction

Generative models related to audio have seen significant improvements due to advances in diffusion models, autoregressive transformers, and representation learning. These systems can now generate music which is closely similar to human produced compositions. While these models offer many benefits in terms of creativity and production of music, they also introduce new challenges with regards to verifying the authenticity of musical content. The detection of AI generated music has widespread effects to domains like digital authentication, rights management, and forensic analysis.

Existing research in audio forensics has explored spectral irregularities, statistical artefacts, and deep-learning classifiers to distinguish real from synthetic content. However, most prior work is limited to speech, and relatively few studies analyse AI vs human discrimination in the music domain, where temporal structure, instrumentation, and production effects are more complex.

This paper addresses this gap by comparatively evaluating several deep audio models on a dataset consisting of both human and AI generated music. Our contributions include:

- Development of a standardised audio preprocessing pipeline and unified spectrogram representation.
- Implementation and evaluation of four model families: CNN, AE classifier, transformer-based classifier, and hybrid fusion classifier.
- A comparative analysis framework using quantitative metrics and visualisations.
- A discussion of dataset bias, generalisation challenges, and future research directions.

The rest of the paper is organised as follows: Section 2 reviews related literature; Section 3 details the methodology; Section 4 explains the experimental setup; Section 5 presents results; Section 6 discusses limitations and future work; Section 7 concludes the paper.

2 Related Work

Only lately have a few studies started addressing the detection of entirely AI generated music, whereas a significant body of research over the previous few years has focused on the detection of synthetic audio, primarily voice. In order to provide baselines and architectural insights for musical deepfake detection, we evaluate pertinent work in three main areas: (1) audio deepfake and synthetic speech detection; (2) early attempts to detect AI generated music; and (3) general audio categorization and music tagging.

2.1 Audio Deepfake and Synthetic Speech Detection

Synthetic speech detection has been the subject of a large amount of previous research. A thorough overview of deepfake speech production, detection pipelines, time-frequency features, and robustness difficulties can be found in Khanjani et al.’s study [7]. Lam et al. [9] investigate spectrogram representations such as STFT, CQT, and Wavelet transforms, assess CNN-, RNN-, and hybrid-based classifiers, and report a low equal error rate (0.03) on ASVspoof 2019 utilizing ensemble learning.

Additionally successful are hybrid temporal spectral designs. On the Fake-or-Real (FoR) dataset, a CNN LSTM model assessed in [14] achieves 94.7% accuracy while maintaining high cross-dataset performance. The new survey by [13] provides a more comprehensive review of audio deepfake detection techniques,

highlighting difficulties with generalization across languages and acoustic environments. The parallel LSTM autoencoder system presented in [15], which combines MFCC, wavelet, prosodic, and glottal features and produces ROC-AUC values close to 98%, exhibits more sophisticated temporal modeling.

There have also been significant benchmark initiatives. The ASVspoof 2021 challenge [18] revealed substantial generalization gaps in existing detectors by standardizing evaluation methodologies for synthetic and replay assaults. In a similar vein, the ADD 2023 audio deepfake challenge [19] showed that cross corpus generalization is difficult for models trained on small domains. Additionally, research on only audio detection is further motivated by multimodal deepfake datasets like FakeAVCeleb [6], which show the growing prevalence of audiovisual synthesis.

These studies’ methodological insights feature design, spectrum modeling, and robustness evaluation form a crucial basis for musical deepfake detection even though they concentrate on speech rather than music.

2.2 Detecting AI Generated Music

Deepfake detection of music is a very young field. One of the first extensive resources with complete AI generated songs (vocals, lyrics, and instruments) alongside actual tracks is the SONICS dataset, which was initially presented in [11]. When used on whole musical compositions, it illustrates the shortcomings of previous singing-voice deepfake systems.

In [12], a more targeted classification strategy is investigated using the FakeMusicCaps dataset to train CNN classifiers based on mel-spectrograms. Many detectors find it difficult to generalize when musical alterations are introduced, according to their augmentation trials (pitch shifting, tempo variations). The Segment Transformer suggested in [8] offers a structural modeling viewpoint by segmenting audio and modeling inter-segment relationships to examine long-range musical patterns. Strong results on SONICS and FakeMusicCaps are obtained with this method, emphasizing the significance of structure cues as opposed to only spectral ones.

Additionally, Zhang et al. demonstrate that localization-based cues can supplement global classification by introducing MUSDL [20], a system for localizing manipulations in music deepfakes. The scope and complexity of contemporary music generation models are demonstrated by foundational generative systems like OpenAI Jukebox [3], highlighting the necessity of detection techniques that take long range musical structure into account.

According to Li et al.’s conceptual overview [10], AI generated music detection necessitates essentially different approaches than speech deepfake detection because of its longer temporal structure, greater musical complexity, genre diversity, and instrumentation layering.

When taken as a whole, these studies represent the beginning of a new line of inquiry focused on identifying AI generated music, which is different from speech-oriented deepfake detection.

2.3 Music Classification and Tagging as Baselines

A significant body of work on music tagging, genre classification, and audio event detection emerged prior to the popularity of AI generated music. Strong baselines for modeling musical sounds are provided by these pieces. With proper data augmentation (pitch shifting, temporal stretching, and noise injection), deep neural networks trained on Mel spectrograms such as the CSN model in [16] can achieve nearly perfect accuracy (99.9%) on datasets like GTZAN.

For tasks like music tagging, mood categorization, and instrument recognition, traditional CNN models that use spectrograms or MFCCs have long been successful. Since robust spectral and temporal representations are also necessary for musical deepfake detection, these architectures continue to be highly significant.

2.4 Gaps and Challenges

Despite advancements, there are still large gaps. Despite the fact that music has more complicated structures and complex instrumentation, the majority of research on audio deepfake detection concentrates on speech rather than music. As generative models advance, the reliance of existing music deepfake detectors on spectral anomalies may decrease. Although promising, datasets like SONICS and FakeMusicCaps are still constrained in terms of real-world variability and genre diversity.

Modeling a broad temporal structure, resilience to changes in pitch and tempo, and invariance to practical effects like compression, mixing, or remastering are all necessary for identifying AI generated music. These findings highlight the necessity of doing a thorough assessment of contemporary deep audio models, such as CNNs, audio transformers, and hybrid architectures, in order to identify music produced by artificial intelligence.

3 Methodology

This section describes the proposed end-to-end pipeline, including dataset preparation, preprocessing, representation learning, and the four classification models under consideration.

3.1 Dataset Preparation

Our dataset includes a mix of human composed music and AI generated pieces, ensuring a diverse range of musical styles and generation methods.

Human Music Dataset The music samples in this dataset were retrieved from the Free Music Archive (FMA) dataset, specifically the `fma_small` subset. The following steps were carried out in the preparation pipeline:

1. **Data Acquisition:** Downloaded the FMA small dataset (~8,000 tracks, 30-second clips).
2. **Random Sampling:** Selected 4,000 MP3 files at random from the dataset to eliminate genre bias.
3. **Format Conversion:** Converted all MP3 files to WAV format using `ffmpeg` and `pydub`, with resampling to 16 kHz and conversion to mono audio.
4. **Duration Normalization:** Trimmed each audio clip to exactly 5 seconds using `librosa`, keeping only the initial segment to maintain consistency.
5. **Packaging:** Compressed the processed dataset into a single archive for distribution and reproducibility.

AI Generated Music Dataset The AI generated samples were sourced from two datasets to ensure diversity in generation techniques:

1. **MusicGen Samples (2,000 clips):** Generated music ourselves using Meta’s MusicGen model [2] via the Hugging Face Transformers library [17]. The generation process involved:
 - Randomly sampling prompts combining genres (electronic, jazz, classical, etc.), moods (dark, uplifting, calm, etc.), and instruments (piano, synths, guitar, etc.).
 - Generating 5-second clips (256 tokens) using the `facebook/musicgen-small` model with batch processing for efficiency.
 - Storing metadata including generation prompts for each sample.
2. **AISHELL-7A Dataset:** Supplemented with samples from the AISHELL-7A synthetic audio dataset [1], which provides additional diversity in AI generation architectures and musical characteristics.

The final dataset contains approximately 4,000 human samples and 4,700 AI generated samples. This ensures that the datasets are balanced.

3.2 Audio Preprocessing

Following the dataset preparation pipeline, we applied a similar pipeline to all audio clips to standardise them and ensure uniformity in the input representations. Since resampling and duration normalisation was carried out previously, this pipeline focuses on feature extraction:

1. **Waveform Loading:** Audio files are loaded as mono waveforms at 16 kHz sampling rate.
2. **Mel-Spectrogram Extraction:** From each waveform, we compute a mel-spectrogram with $M = 128$ mel bands using a short-time Fourier transform (STFT) with window size $N_{\text{FFT}} = 2048$ and hop length $H = 512$.
3. **Log-Magnitude Conversion:** The power spectrogram is converted to decibel (dB) scale using:

$$X_{\text{dB}} = 10 \log_{10}(\text{mel-spectrogram} + \epsilon),$$

where ϵ is a small constant to avoid numerical instability.

4. **Normalization:** Each spectrogram is normalized to the range $[0, 1]$ by scaling relative to the maximum value.

The resulting normalized mel-spectrogram is stored as a tensor

$$X \in \mathbb{R}^{1 \times M \times T},$$

where $M = 128$ mel bands and T represents the number of time frames (approximately 157 frames for 5-second clips). This representation serves as the common input across all classification models.

3.3 Model Families Evaluated

We worked with four different models: a convolutional neural network operating directly on spectrograms, an autoencoder-based representation classifier to highlight deviations from natural human audio, a transformer-based audio representation model to capture long-range musical context via self-attention, and a hybrid fusion model that combines autoencoder and transformer embeddings to test whether combining these results in stronger discrimination between human and AI generated music.

Convolutional Neural Network (CNN) The CNN ingests the spectrogram X and processes it through multiple convolutional blocks, each consisting of a convolution, batch normalisation, non-linear activation, and pooling operation. Let $H_0 = X$ and for each layer l we compute

$$H_l = f_l(H_{l-1}) = \sigma(\text{BN}(W_l * H_{l-1} + b_l)),$$

where W_l and b_l are the convolutional weights and biases, BN denotes batch normalisation, σ is a rectified linear unit (ReLU), and $*$ denotes convolution. After several such layers and spatial pooling operations, the final feature map is flattened and passed through fully connected layers to produce logits

$$y = \text{FC}(H_L),$$

where L is the number of convolutional layers. The model is trained using cross-entropy loss on the binary labels (AI vs human music).

Multi-Task Autoencoder Representation and MLP Classifier The second model family uses a multi-task convolutional autoencoder trained on both human and AI generated music to learn a compact latent representation. Unlike traditional autoencoders trained solely on human music, this multi-task architecture simultaneously optimizes for two objectives:

1. **Reconstruction loss:** Learning general musical features through spectrogram reconstruction

2. **Classification loss:** Learning discriminative features for AI vs. human detection

The encoder $E(\cdot)$ maps the spectrogram X to a latent embedding

$$z_{\text{AE}} = E(X) \in \mathbb{R}^d,$$

where $d = 256$ is the latent dimensionality. The decoder $D(\cdot)$ reconstructs the spectrogram:

$$\hat{X} = D(z_{\text{AE}}).$$

The autoencoder is trained with a combined loss function:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}},$$

where $\mathcal{L}_{\text{rec}} = \|X - \hat{X}\|_2^2$ is the reconstruction loss, $\mathcal{L}_{\text{cls}}$ is the cross-entropy classification loss, and $\lambda_{\text{rec}} = 1.0$, $\lambda_{\text{cls}} = 0.5$ are weighting coefficients.

Once trained, the encoder is frozen and used as a feature extractor. A separate multilayer perceptron (MLP) classifier receives z_{AE} as input for the baseline evaluation.

Audio Spectrogram Transformer (AST) Model The third model family employs the Audio Spectrogram Transformer (AST) [5], a Vision Transformer (ViT) architecture adapted for audio spectrograms. AST is pre-trained on AudioSet [4], a large-scale dataset containing 10 million audio samples across 527 audio event classes.

The architecture treats mel-spectrograms as images and divides them into 16×16 patches. Each patch is linearly embedded and fed into a transformer encoder consisting of 12 layers with multi-head self-attention. The scaled dot-product attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

where Q , K , and V are the query, key, and value matrices and d_k is the key dimensionality.

The model extracts a 768-dimensional embedding from the [CLS] token, which is then projected to a 512-dimensional representation:

$$z_{\text{T}} = \text{Proj}(\text{AST}(X)) \in \mathbb{R}^{512}.$$

For our task, the pre-trained AST backbone is frozen, and only the projection layer and classification head are fine-tuned on the AI vs. human music detection task using cross-entropy loss.

Hybrid Fusion Model with Attention Mechanism The hybrid model combines the complementary strengths of reconstruction-based (AE) and semantic (Transformer) representations through an **attention-based fusion mechanism**. Unlike simple concatenation, this approach dynamically weights the importance of each feature source.

First, the autoencoder latent vector and transformer embedding are concatenated:

$$z_{\text{concat}} = [z_{\text{AE}} \parallel z_{\text{T}}] \in \mathbb{R}^{768},$$

where $\parallel$ denotes vector concatenation ($256 + 512 = 768$ dimensions).

An attention module then computes dynamic weights for each feature type:

$$\alpha_{\text{AE}}, \alpha_{\text{T}} = \text{softmax}(\text{Attention}(z_{\text{AE}}, z_{\text{T}})),$$

where α_{AE} and α_{T} represent the learned importance of AE and Transformer features, respectively.

The attention-weighted features are then fused:

$$z_{\text{hybrid}} = \alpha_{\text{AE}} \cdot z_{\text{AE}} \parallel \alpha_{\text{T}} \cdot z_{\text{T}}.$$

This fused representation is fed into a deep MLP classifier with four fully connected layers, batch normalization, ReLU activations, and dropout (rates: 0.5, 0.4, 0.3) for regularization. The classifier is trained with cross-entropy loss while the encoder backbones remain frozen.

Architectural Details:

- Autoencoder: 4-layer CNN encoder/decoder, 256-dim latent space
- AST Transformer: 12 layers, 12 attention heads per layer, 768-dim hidden state, 512-dim output embedding
- Hybrid MLP: $768 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 2$ (with batch norm and dropout)

4 Experimental Setup

4.1 Datasets

The dataset used in this study is made up of both human and AI generated music from various sources. Human-composed audio is extracted from a carefully chosen portion of the FMA dataset, whilst AI generated samples are derived using models like MusicGen and MusicEval. Every audio file has metadata, such as a track number, source dataset, and label (human or AI).

We use grouped splitting based on track identities to avoid data leaking between training, validation, and test splits. Even if several segments are produced from the same track, this guarantees that no track occurs in more than one split. Roughly 70% of the songs are allocated to the training set, 15% to the validation set, and 15% to the test set.

4.2 Training Configuration

Every model is implemented using PyTorch. The CNN and MLP classifiers are trained using the Adam optimiser with the appropriate learning rate and mini-batch size (to be determined precisely based on the final experimental design). To lessen overfitting, early stopping based on validation accuracy or validation loss is used. Before moving on to the classification stage, when AE characteristics are retrieved for every sample, the autoencoder is trained solely on human music.

Note: After the last experimental runs are finished, the precise hyperparameters (learning rates, batch sizes, number of epochs, weight decay, and dropout rates) should be entered here. In order for these values to be updated consistently with the final results, the current language purposefully leaves them implicit.

4.3 Evaluation Metrics

We evaluate model performance using several complementary metrics:

- **Accuracy:** The proportion of correctly classified samples.
- **Precision:** The fraction of predicted AI generated samples that are truly AI generated.
- **Recall:** The fraction of actual AI generated samples that are correctly identified.
- **F1-score:** The harmonic mean of precision and recall.
- **ROC-AUC:** The area under the receiver operating characteristic curve, measuring the trade-off between true positive and false positive rates across thresholds.

Furthermore, we use dimensionality reduction (e.g., t-SNE) to visualize the structure of the learnt embedding spaces and produce confusion matrices and ROC curves for qualitative analysis.

5 Results

5.1 Quantitative Comparison

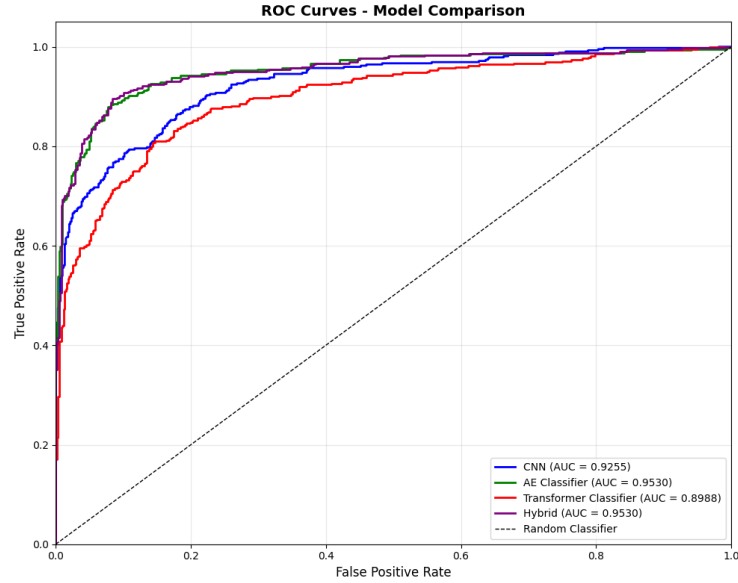
Table 1 reports the performance of all four model families on the held-out test set. For the CNN, AE-based classifier, and hybrid model, the entries may be populated using the latest completed experiments. For the transformer-based model, placeholders are included and must be updated once the corrected implementation and final training runs are available.

Table 1. Performance comparison of all models on the test set. Values for the transformer-based model should be filled in after final experiments.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
CNN (spectrogram)	85.21%	0.8668	0.7577	0.8086	0.9255
AE + MLP	87.86%	0.9459	0.7482	0.8355	0.9530
Transformer + MLP	82.66%	0.8262	0.7340	0.7774	0.8988
Hybrid (AE + Transformer)	86.68%	0.9642	0.7031	0.8132	0.9530

5.2 Visual Analysis

Figures 1, 3 provide qualitative insight into the classifier behaviour. ROC curves illustrate the trade off between sensitivity and specificity for each model, while confusion matrices highlight systematic error patterns. Embedding visualisations obtained via t-SNE suggest how well AI generated and human music separate in the learned feature spaces.

**Fig. 1.** ROC curves comparing different models.

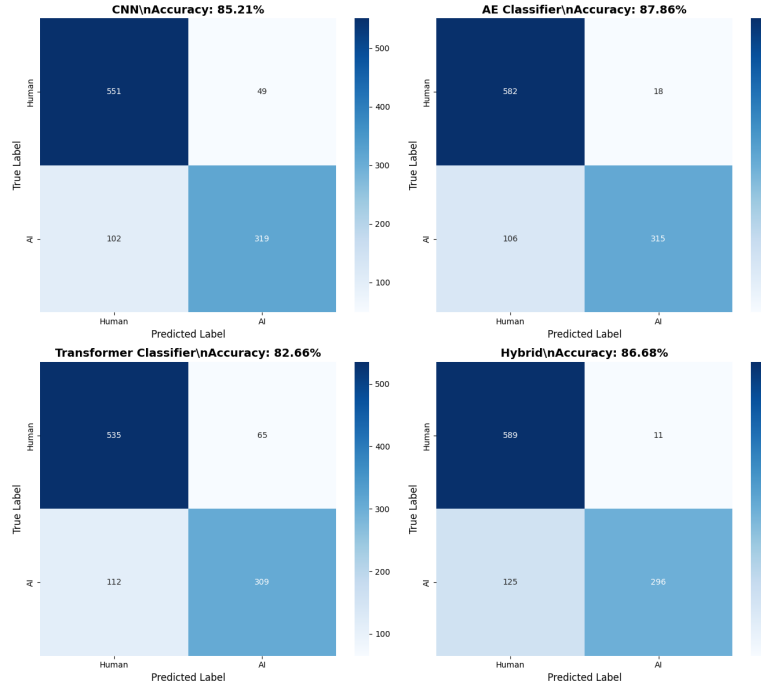


Fig. 2. Confusion matrices for all models.

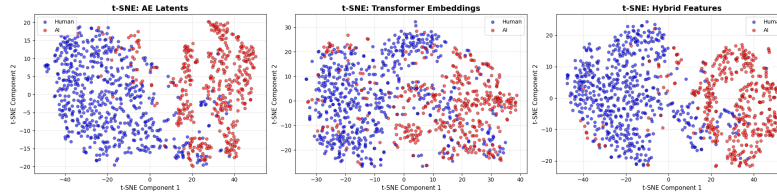


Fig. 3. t-SNE visualisations of learned embeddings.

6 Discussion, Limitations, and Future Work

The purpose of the comparative evaluation framework created in this paper is to clarify the ways in which various audio representation techniques impact the effectiveness of AI generated music detection. Given their capacity to capture local time frequency patterns, CNN-based models that operate directly on spectrograms are anticipated to produce robust baselines. Reconstruction-oriented elements are encoded via autoencoder based representations, which may draw

attention to variations from the variety of human music. When properly constructed and trained, transformer based models are expected to represent global musical context and longer range temporal structure.

There are a few restrictions to be aware of. First, there may be biases unique to this dataset because it blends several AI generators with a single core human music source. Rather than relying just on semantic or stylistic cues, a classifier may take advantage of variations in production quality or encoding methods. Second, very big models are limited by the dataset’s small size, which may necessitate thorough regularization and data augmentation. Third, the transformer based model is still being developed for this project; consistent results are required before any inferences about its performance and advantages over CNN baselines can be made.

Future research should study robustness to adverse or purposefully veiled AI generated music, perform cross-dataset evaluation (training on one subset and testing on another), and expand the dataset to include different human and AI sources. According to the project bonus requirements, an expanded version of this work should also include ablation investigations with other hyperparameters, representation choices, and training protocols.

7 Conclusion

This research describes the development of a comparison framework for detecting AI-generated music with deep auditory models. We proposed a uniform preprocessing pipeline, detailed four complimentary model families (CNN, autoencoder-based classifier, transformer-based model, and hybrid fusion model), and established a multi-metric evaluation technique. While certain quantitative results for the transformer-based model are still pending, the study’s structure is in place to allow for a thorough quantitative assessment once all experiments are completed.

This work is significant because it provides a systematic framework for comparing audio representations and architectures in the context of AI music recognition, a topic that will only grow in importance as generative models become more widespread. This paper, by clearly highlighting limits and describing future enhancements, is intended to serve as a solid basis prospective future research on robust AI generated music recognition.

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